ANALYSIS I — SHEET 3 (for Week 5 tutorials)
Sequences, Convergence, Boundedness, Convergence to Infinity.

[You may assume standard properties of sin and cos in Questions 1 and 2, even though they have not been rigorously defined yet in the course.]

1. For each of the following sequences (a_n), and for arbitrary $\varepsilon > 0$, find N such that $|a_n| < \varepsilon$ whenever $n \geqslant N$. [*Note: you only need to find a value of N that works and not necessarily the smallest such N.*]

$$\frac{1}{n^2}; \qquad \frac{1}{n(n-\pi)}; \qquad \frac{\sin(n^2)}{2^n}; \qquad \sin\frac{1}{n}.$$

2. For $n \geqslant 1$ let

$$a_n = 1 + \frac{2^{n/1000} - 1}{n^2} \quad \text{and} \quad b_n = \frac{n^2 \cos n}{2^{n/1000} - 1}.$$

Use a calculator to determine a_n and b_n for $n = 1, 2, 5, 10, 100, 1000$. What do you think is the behaviour of a_n and b_n as n tends to infinity?

3. Use the binomial theorem to show that

$$\left(1 + \frac{2}{\sqrt{n}}\right)^n \geqslant n \quad \text{for } n \geqslant 1,$$

and deduce that $n^{1/n} \to 1$ as $n \to \infty$.

4. (i) Suppose that (a_n) is bounded and $b_n \to 0$ as $n \to \infty$. Prove that $a_n b_n \to 0$.

(ii) Suppose that $a_n \to \infty$ and $b_n \to L$ as $n \to \infty$, where L is a real number. Prove that $b_n/a_n \to 0$ and $a_n + b_n \to \infty$ as $n \to \infty$.

5. (i) Write down, carefully quantified, what it means for a real sequence (a_n) (a) to be convergent; (b) to be bounded; (c) to tend to infinity.

(ii) Write down, carefully quantified, the negations of (a), (b), (c).

(iii) Prove carefully that a convergent sequence is bounded. Give an example of a bounded divergent sequence.

(iv) Prove carefully that a sequence which tends to infinity is unbounded. Give an example of an unbounded sequence which does not tend to infinity.

(v) Prove that a convergent real sequence has a maximum or a minimum (or both). Is the same true of bounded real sequences?

6. Let c be a complex number. The sequence $z_n(c)$ is defined recursively by

$$z_1(c) = c, \qquad z_{n+1}(c) = z_n(c)^2 + c \quad \text{for} \quad n \geqslant 1.$$

The *Mandelbrot Set* is defined by $M = \{c \in \mathbb{C} \mid \text{ the sequence } z_n(c) \text{ is bounded}\}$.

(i) Show that $-2, -1, 0, i$ each lie in M but that $1 \notin M$.

(ii) Show that if $c \in M$ then $\bar{c} \in M$, where $\bar{c}$ denotes the conjugate of c.

(iii) Show that if $|c| \leqslant 1/4$ then $|z_n(c)| < 1/2$ for all n. (So if $|c| \leqslant 1/4$ then $c \in M$.)

(iv) Show that if $|c| = 2+\varepsilon$ where $\varepsilon > 0$, then $|z_n(c)| \geqslant 2 + a_n\varepsilon$ for $n \geqslant 1$ where $a_n = (4^n + 2)/6$. Deduce that the Mandelbrot lies entirely within the disc $|z| \leqslant 2$.

Extension Exercises (Optional)

7. A *metric* on a set X is a map $d\colon X^2 \to \mathbb{R}$ which satisfies properties M1, M2, M3 of Definition 1.44. Show that the following maps d_1, d_2, d_∞ are metrics on $\mathbb{R}^2$.

(i) $d_1\left((x_1,y_1),(x_2,y_2)\right) = |x_1 - x_2| + |y_1 - y_2|$.

(ii) $d_2\left((x_1,y_1),(x_2,y_2)\right) = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$.

(iii) $d_\infty\left((x_1,y_1),(x_2,y_2)\right) = \max(|x_1 - x_2|, |y_1 - y_2|)$.

8. Given a metric d on a set X, we say that a sequence (x_n) in X is convergent if there exists $L \in X$ such that $d(x_n, L) \to 0$ as $n \to \infty$.

(i) Show that the limit of a sequence, if it exists, is unique.

(ii) For each metric d_1, d_2, d_∞, sketch the set of points which are distance 1 from the origin.

(iii) Given a sequence $(\mathbf{x}_n)$ in $\mathbb{R}^2$, show that if $(\mathbf{x}_n)$ converges to $\mathbf{L}$ using one of the metrics d_1, d_2, d_∞ then it converges to $\mathbf{L}$ using each of them.

9. (i) (*Dirichlet's Approximation Theorem*) Let α be an irrational number. By considering which of the intervals $[k/n, (k+1)/n]$, where $0 \leqslant k \leqslant n-1$, the fractional parts of $0, \alpha, 2\alpha, \ldots, n\alpha$ fall into, show that there exist integers p, q with $0 < q \leqslant n$ such that $|q\alpha - p| < 1/n$.

(ii) Given real numbers $a < b$, show that there are integers m, n such that $a < m + n\alpha < b$.

10. The *hyperreals* ${}^*\mathbb{R}$ are an extension of the real numbers that allow for the rigorous treatment of infinitesimals and infinities. The definition relies on the use of a map $m\colon \mathcal{P}(\mathbb{N}) \to \{0,1\}$ defined on subsets of $\mathbb{N}$ with the properties:

(I) $m(\mathbb{N}) = 1$ and $m(A) = 0$ for a finite set A.

(II) if A, B are disjoint subsets of $\mathbb{N}$ then $m(A \cup B) = m(A) + m(B)$.

A hyperreal x is represented by a sequence (x_n) and two real sequences (x_n) and (y_n) represent the same hyperreal if

$$m\left(\{n \in \mathbb{N} \mid x_n = y_n\}\right) = 1.$$

A real number c is identified with the constant sequence $(c) = (c, c, c, \ldots)$.

(i) Addition and multiplication of hypereals is done termwise. Explain why the product

$$(1,0,1,0,1,\ldots) \times (0,1,0,1,0,\ldots) = (0,0,0,0,0,\ldots)$$

is equivalent to one of the products $1 \times 0 = 0$ or $0 \times 1 = 0$ depending on the particular m used in the construction of ${}^*\mathbb{R}$.

(ii) Show that a non-zero hyperreal has a multiplicative inverse.

(iii) The hyperreals can be ordered by setting

$$(x_n) \leqslant (y_n) \quad \Longleftrightarrow \quad m\left(\{n \in \mathbb{N} \mid x_n \leqslant y_n\}\right) = 1.$$

Show that $\leqslant$ is a total order. [*Hint: you will need to show if* $m(S) = m(T) = 1$ *then* $m(S \cap T) = 1$.]

11.(i) A hyperreal (x_n) is said to be *infinitesimal* if $(-\varepsilon) < (x_n) < (\varepsilon)$ for all positive ε. Show that if $x_n \to 0$ then (x_n) represents an infinitesimal hyperreal. Show the converse is not true.

(ii) A hyperreal (x_n) is said to be *finite* if $(-a) < (x_n) < (a)$ for some positive a. Show that a finite hyperreal (x_n) can be uniquely written as $(x_n) = (x) + (y_n)$ where x is real and (y_n) is infinitesimal.

(iii) A hyperreal is said to be *infinite* if it is not finite. Show that (n) is an infinite hyperreal.

(iv) Show that the set of infinitesimals is non-empty and bounded above, but has no supremum. Hence ${}^*\mathbb{R}$ is not complete.